

```
> #Programmer Name: Peter Chapman
> #Date: November 5, 2024
> #This code reads in the Car Mileage Data Set. It then finds
> #the percentiles of miles per gallon, regardless of country.
> #Then it finds the ranks and creates a new data frame with ranks.
> #Summary values are then found using this data frame. Finally,
> #the new data frame is used to find percentiles and summaries
> #for the miles per gallon of both countries separately.
>
```

```
> #reading in CSV data set to data frame
```

```
> mpg<-read.csv("Car Mileage.csv")
```

```
> mpg
```

```
Country MPG
```

```
1 United States 18
2 United States 15
3 United States 18
4 United States 16
5 United States 17
6 United States 15
7 United States 14
8 United States 14
9 United States 14
10 United States 15
11 United States 15
12 United States 14
13 United States 15
14 United States 14
15 United States 22
16 United States 18
```

17 United States 21
18 United States 21
19 United States 10
20 United States 10
21 United States 11
22 United States 9
23 United States 28
24 United States 25
25 United States 19
26 United States 16
27 United States 17
28 United States 19
29 Japan 24
30 Japan 27
31 Japan 27
32 Japan 25
33 Japan 31
34 Japan 35
35 Japan 24
36 Japan 19
37 Japan 28
38 Japan 23
39 Japan 27
40 Japan 20
41 Japan 22
42 Japan 18
43 Japan 20
44 Japan 31
45 Japan 32

```

46   Japan 31
47   Japan 32
48   Japan 24
49   Japan 26
50   Japan 29
51   Japan 24
52   Japan 24
53   Japan 33
54   Japan 33
55   Japan 32
56   Japan 28
>
> #0, 25th, 50th, 75th, and 100th percentiles of mpg,
> #regardless of country
> mpg_Percentiles<-quantile(mpg$MPG)
> mpg_Percentiles
  0%  25%  50%  75% 100%
9.00 15.75 21.00 27.00 35.00
>
> #minimum miles per gallon, regardless of data set
> mpg_min<-min(mpg$MPG)
> mpg_min
[1] 9
>
> #maximum miles per gallon, regardless of data set
> mpg_max<-max(mpg$MPG)
> mpg_max
[1] 35
>

```

```
> #Calculating ranks, creating new data frame
```

```
> Ranks<-rank(mpg$MPG)
```

```
> mpg_with_ranks<-data.frame(mpg, Ranks)
```

```
> mpg_with_ranks
```

```
Country MPG Ranks
```

```
1 United States 18 20.5
2 United States 15 12.0
3 United States 18 20.5
4 United States 16 15.5
5 United States 17 17.5
6 United States 15 12.0
7 United States 14 7.0
8 United States 14 7.0
9 United States 14 7.0
10 United States 15 12.0
11 United States 15 12.0
12 United States 14 7.0
13 United States 15 12.0
14 United States 14 7.0
15 United States 22 30.5
16 United States 18 20.5
17 United States 21 28.5
18 United States 21 28.5
19 United States 10 2.5
20 United States 10 2.5
21 United States 11 4.0
22 United States 9 1.0
23 United States 28 45.0
24 United States 25 38.5
```

25 United States 19 24.0

26 United States 16 15.5

27 United States 17 17.5

28 United States 19 24.0

29 Japan 24 35.0

30 Japan 27 42.0

31 Japan 27 42.0

32 Japan 25 38.5

33 Japan 31 49.0

34 Japan 35 56.0

35 Japan 24 35.0

36 Japan 19 24.0

37 Japan 28 45.0

38 Japan 23 32.0

39 Japan 27 42.0

40 Japan 20 26.5

41 Japan 22 30.5

42 Japan 18 20.5

43 Japan 20 26.5

44 Japan 31 49.0

45 Japan 32 52.0

46 Japan 31 49.0

47 Japan 32 52.0

48 Japan 24 35.0

49 Japan 26 40.0

50 Japan 29 47.0

51 Japan 24 35.0

52 Japan 24 35.0

53 Japan 33 54.5

```

54    Japan 33 54.5
55    Japan 32 52.0
56    Japan 28 45.0
>
> #Summary values of mpg_with_ranks data frame
> Summary_mpg_all<-summary(mpg_with_ranks$MPG)
> Summary_mpg_all
  Min. 1st Qu.  Median   Mean 3rd Qu.   Max.
 9.00 15.75 21.00 21.59 27.00 35.00
>
> #US percentiles using mpg_with_ranks data frame
> US_Percentiles<-with(mpg_with_ranks, quantile(MPG[Country=="United States"]))
> US_Percentiles
 0% 25% 50% 75% 100%
9.00 14.00 15.50 18.25 28.00
>
> #Japan percentiles using mpg_with_ranks data frame
> JP_Percentiles<-with(mpg_with_ranks, quantile(MPG[Country=="Japan"]))
> JP_Percentiles
 0% 25% 50% 75% 100%
18 24 27 31 35
>
> #US summary using mpg_with_ranks data frame
> US_Summary<-with(mpg_with_ranks, summary(MPG[Country=="United States"]))
> US_Summary
  Min. 1st Qu.  Median   Mean 3rd Qu.   Max.
 9.00 14.00 15.50 16.43 18.25 28.00
>
> #Japan summary using mpg_with_ranks data frame

```

```
> JP_Summary<-with(mpg_with_ranks, summary(MPG[Country=="Japan"]))
```

```
> JP_Summary
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
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18.00	24.00	27.00	26.75	31.00	35.00
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